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Cognitive reserve moderates the functional connectivity changes in resting-state

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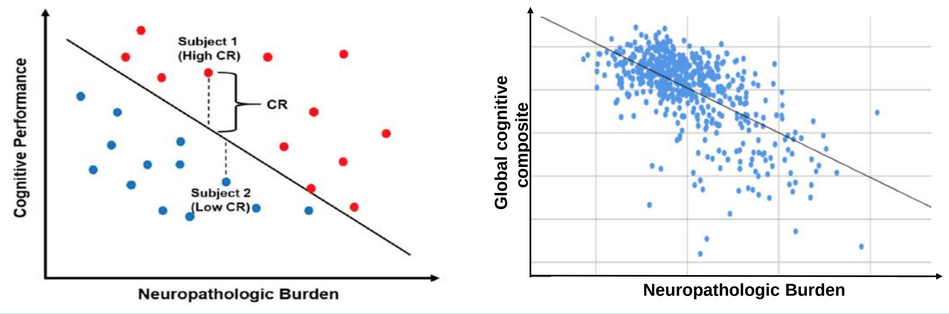
Background: Cognitive reserve (CR) refers to adaptability of cognitive processes that can explain the individual differences in cognition relative to brain aging or neurodegenerative changes e.g. Alzheimer's disease (AD)[1]. Recently, various studies tried to quantify a marker that represents CR. Among others, the residual approach suggested being relatively reliable as it has been studied not only with cross-sectional but also with longitudinal observations[2, 3]. However, it is unknown how and which changes in functional network structures in resting-state are related to cognitive reserve.

Methods: In the present study we defined a CR Marker based on residuals calculated through a multimodal (multi-linear) regression model adapted from a recent study[3] cortical thickness of vulnerable regions and hippocampal volume derived by using magnetic resonance imaging (MRI), as well as genetic and sociodemographic information as predictors (**Fig. 1**). The data of 818 participants including healthy controls and with subjective cognitive decline (SCD), patients with mild cognitive impairment (MCI) and probable AD from the German multicentric DELCODE cohort was assessed (**Table 1**). To examine neural substrates of defined CR marker, we constructed a weighted undirected partial correlation matrix using the Functional connectivity analysis was performed using the CONN-fMRI Functional Connectivity Toolbox[4] (ver. 17, www.nitrc.org/projects/conn) and SPM 12 (www.fil.ion.ucl.ac.uk/spm/) and analyzed graph theoretical metrics[5] in GraphVar toolbox[6] following the removal of negative weights in global and nodal topography with various cut-off threshold values. Analyses were conducted by using permutation-based statistic. False discovery rate (FDR)[7] for the correction of alpha level is used in the analysis in local topography. We examined the possible associations between CR and diagnosis alterations in functional connectivity. The following list of metrics have been calculated at a global level: (1) *global transitivity*, (2) *global efficiency*, (3) *modularity* under the Louvain method. Then on local level: (1) *local efficiency*, (2) normalized *clustering coefficient*, (3) *betweenness centrality*, and (4) *degree*. The analyses were conducted both in entire network and within default-mode network (DMN) by separating the belonging nodes according to resting-state networks[8] defined in Brainnetome-Atlas[9] (BN-Atlas).

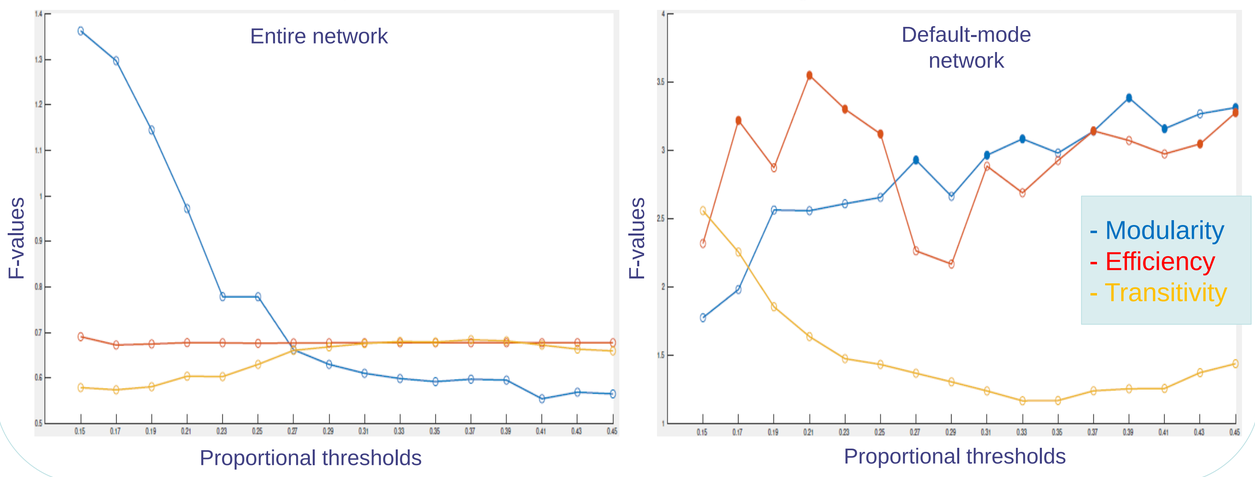
Figure 1: Multivariate regression model

$$\text{Cognition}_{estimated} = \beta_0 + \beta_1 \times X_{Thickness} \cdot \beta_2 \times X_{HCvolume} + \beta_3 \times X_{ApoE} + \beta_3 \times X_{Age} \cdot \beta_4 \times X_{Sex} + \beta_5 \times X_{TIV}$$



Results: CR Marker was positively correlated (r=0.35, p<0.001) with higher years of formal education when adjusted for age, gender and APOE ε4 allele-status. In the graph-theory based analysis of connectivity changes in resting-state-functional-MRI using a region of interest (ROI) based approach by utilizing parcellations of the BN-Atlas, we demonstrated that CR marker is significantly interacting with the diagnosis in a general linear model with the graph measures of *global efficiency (p<0.05)* and *modularity (p<0.05)* of DMN (**Table 2, Fig. 2**). We adjusted the models for age, gender, imaging site, and APOE ε4 allele-status.

Figure 2: F-values of the Interaction term for latent cognitive reserve marker and diagnosis among thresholds (15% to 45%, with 2% increments) in the linear models. Filled rounds represent significant interactions when p<0.05.



	Age[a] (SD)	Years of education[a] (SD)	Gender[b] (N/% female)	ApoE ε4-status[b] (N/% carriers)	CDR-sb[a] (SD)	Global cognitive composite score[a]	Latent CR-Marker[a] (SD)
HC	68.58 (5.43)	14.65 (2.75)	166 (58)	215 (75.2)	0.01 (0.33)	0.47 (0.43)	0.28 (0.48)
SCD	71.13 (6.08)	14.84 (2.94)	141 (44.3)	102 (32.1)	0.36 (0.96)	0.34 (0.53)	0.33 (0.51)
MCI	73.22 (5.83)	13.79 (3.14)	58 (48.3)	57 (47.5)	1.58 (1.29)	-0.6 (0.65)	-0.44 (0.66)
AD	74.94 (6.18)	12.91 (3.08)	56 (59.6)	56 (59.6)	4.49 (2.21)	-1.68 (0.63)	-1.39 (0.75)
p value	**<0.001**	**<0.001**	**0.02**	**<0.001**	**<0.001**	**<0.001**	**<0.001**

Table 1: Demographic and clinical characteristic of the participants and patients CDR-sb, Clinical dementia rating-sum of boxes; CR, cognitive reserve. [a]Kruskal-Wallis-test, [b]Chi-Square-test

On the local level, however, nodes that belong particularly to DMN and cognitive task related networks such as salience, frontoparietal and dorsal attention networks shows a significant interaction (false discovery rate-corrected alpha level <0.05) that are summarized in **Table 3**. In a further analysis on nodal level within DMN, the latent CR-marker and diagnosis interacts in terms of *degree* and *local efficiency in* frontal regions and in terms of *betweenness centrality* in temporal lobe.

These results indicating that CR moderates the effect of the disease stage on network features evaluated by using graph theory both in global and local topographies within DMN. Moreover, the interaction is observable also in nodes belong to other resting-state networks in various measures.

		Entire network			Default mode network		
CR proxy	**Interaction term**	**Global Efficiency**	**Transitivity**	**Modularity**	**Global Efficiency**	**Transitivity**	**Modularity**
CRM	CRM*Diagnosis	-	-	-	p<0.05	-	p<0.05

Table 2: Significant interactions in global graph theoretical measures. Regions are shown when significant (p <0.05). CR, cognitive reserve; CRM, latent CR marker.

Network	Interaction Term	Region	ROI	RSN	Graph measure	P	Thres.
Entire network	CRM*Diagnosis	Left Inferior parietal lobule	A44v	Salience network	Betweenness centrality	<0.001	1
	CRM*Diagnosis	Right inferior parietal lobule	A40rd	Dorsal Attention network	Clustering coefficient	<0.001	1
	CRM*Diagnosis	Right cingulate gyrus	A32sg	Default mode network	Local efficiency	<0.001	1
	CRM*Diagnosis	Right Insular gyrus	vIa	Frontoparietal network	Degree	<0.001	3
Default-mode network	CRM*Diagnosis	Right inferior frontal Gyrus	A45c	Default mode network	Degree	0.001	1
	CRM*Diagnosis	Right inferior frontal Gyrus	A45c	Default mode network	Local efficiency	0.001	1
	CRM*Diagnosis	Right superior temporal sulcus	aSTS	Default mode network	Betweenness centrality	0.001	1

Table 3: Regions with significant interaction in local graph theoretical measures. Regions are shown when FDR-alpha <0.05. CDR-sb, Clinical dementia rating-sum of boxes; CR, cognitive reserve; CRM, latent CR marker; Thres, threshold; ROI abbreviations from BN-Atlas are presented. Number of thresholds are out of 16 thresholds (15% to 45%, with 2% increments) are shown.

Discussion: It was suggested that CR is associated with higher ability of the brain to form more efficient functional networks and optimize the balance between integration and segregation[10]. Our analysis focused on changes in AD by examining the participants in spectrum of AD with individually obtained latent CR marker and we revealed comparable results on local level with previous studies that examined effects of CR on graph theoretical measures[3,10,11]. However, we found interaction in global topography only within DMN in measures of both integrity i.e. *efficiency* and segregation i.e. *modularity*. Considering previous findings, we assume that observed moderation of CR in our study on graph theoretical measures is associated with more efficient networks which is more pronounced in DMN.

These findings can foster understanding of the changes in functional networks in AD and will be useful to help identify individuals with vulnerability or resistance to AD pathology and characterize patients for intervention or drug trials.

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